

11	(a) (i)	e.m. radiation produced whenever charged particle is accelerated electrons hitting target have distribution of accelerations	M1 A1	[2]
	(ii)	<i>either</i> wavelength shorter/shortest for greater/greatest acceleration <i>or</i> $\lambda_{\min} = hc/E_{\max}$ <i>or</i> minimum wavelength for maximum energy all electron energy given up in one collision/converted to single photon	B1 B1	[2]
	(b) (i)	hardness measures the penetration of the beam greater hardness, greater penetration	C1 A1	[2]
	(ii)	controlled by changing the anode voltage higher anode voltage, greater penetration/hardness	C1 A1	[2]
	(c) (i)	long-wavelength radiation more likely to be absorbed in the body/less likely to penetrate through body	B1	[1]
	(ii)	(aluminium) filter/metal foil placed in the X-ray beam	B1	[1]

12 (a) strong uniform (magnetic) field M1